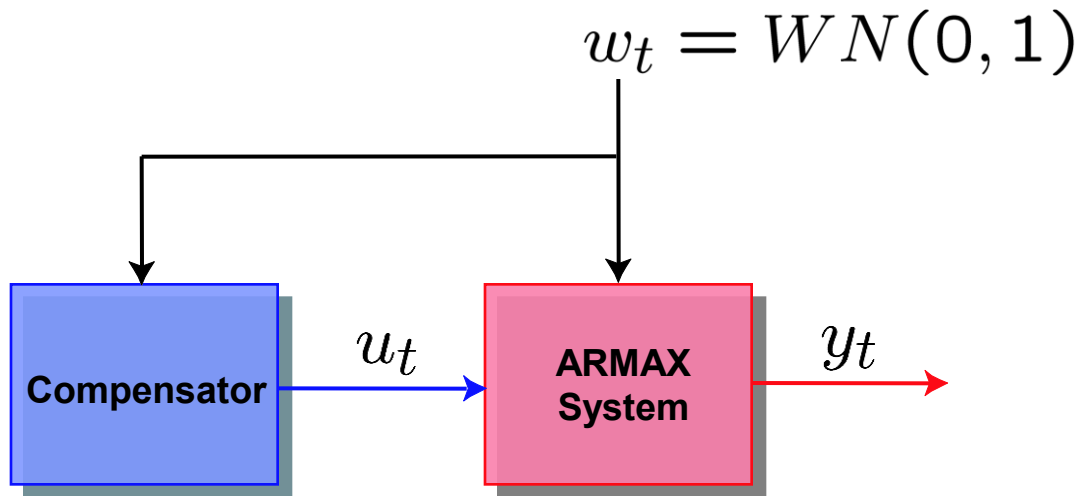
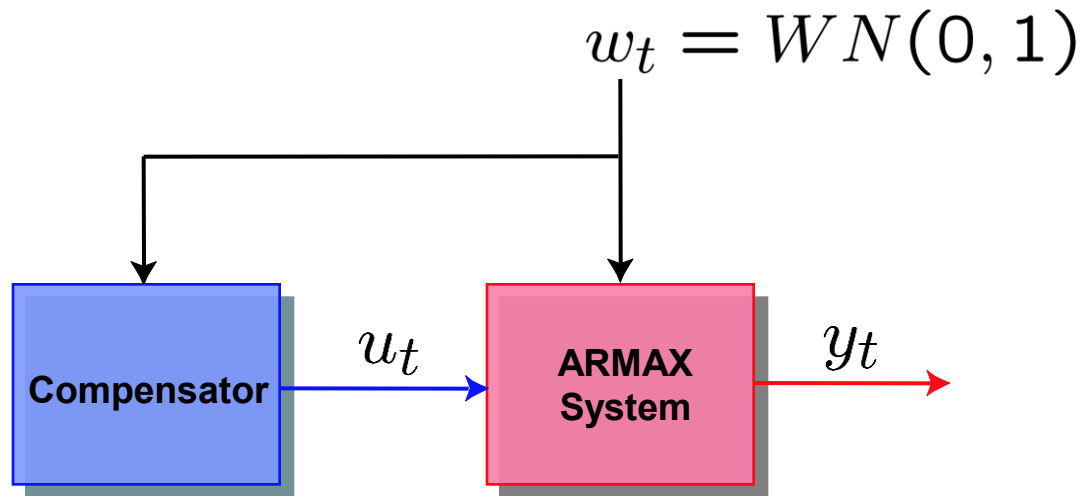


Example: feedforward noise compensation

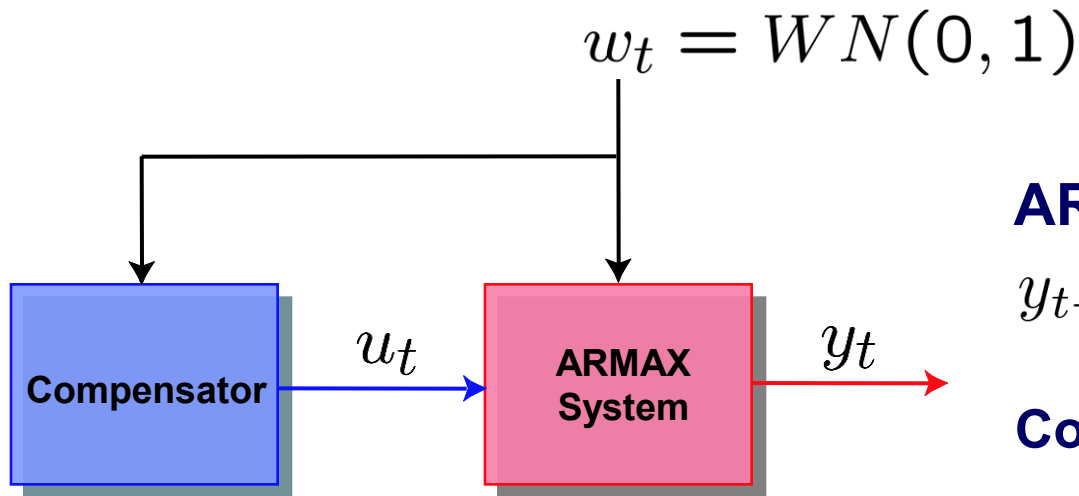


Example: feedforward noise compensation



Objective: reduce the effect of noise on y

Example: feedforward noise compensation



ARMAX System:

$$y_{t+1} = ay_t + bu_t + cw_t + dw_{t-1}$$

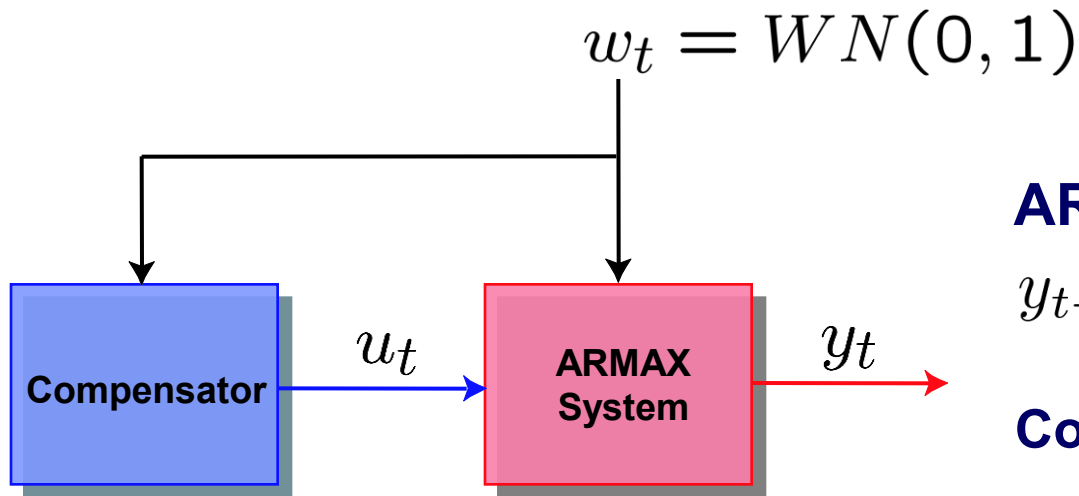
Compensator:

$$u_t = \nu_1 w_t + \nu_2 w_{t-1}$$

$$\delta = [a \ b \ c \ d]^T$$

$$\nu = [\nu_1 \ \nu_2]^T$$

Example: feedforward noise compensation



ARMAX System:

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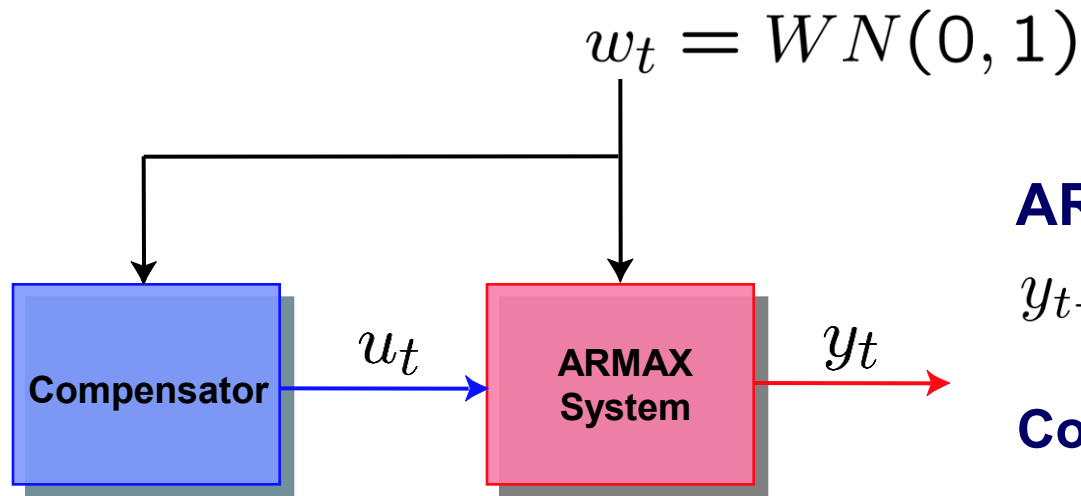
Compensator:

$$u_t = \nu_1 w_t + \nu_2 w_{t-1}$$

Goal:

$$\min \text{var}[y_t]$$

Example: feedforward noise compensation



ARMAX System:

$$y_{t+1} = ay_t + bu_t + cw_t + dw_{t-1}$$

Compensator:

$$u_t = \nu_1 w_t + \nu_2 w_{t-1}$$

$$\text{var}[y_t] = \frac{(c + b\nu_1)^2 + (d + b\nu_2)^2 + 2a(c + b\nu_1)(d + b\nu_2)}{1 - a^2}$$

Example: feedforward noise compensation

system parameters unknown: $a, b, c, d \in \Delta$

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scenario approach:

sample: $a_i, b_i, c_i, d_i \in \Delta, \quad i = 1, 2, \dots, N;$

solve:

$$\min_{\nu_1, \nu_2} \left[\max_i \frac{(c_i + b_i \nu_1)^2 + (d_i + b_i \nu_2)^2 + 2a_i(c_i + b_i \nu_1)(d_i + b_i \nu_2)}{1 - a_i^2} \right]$$

Example: feedforward noise compensation

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Example: feedforward noise compensation

$$\text{var}[y_t] = \frac{(c + b\nu_1)^2 + (d + b\nu_2)^2 + 2a(c + b\nu_1)(d + b\nu_2)}{1 - a^2}$$

$$\Delta = \{a, b, c, d : a = \frac{3.5\sigma_1^2 - 0.2}{3\sigma_1^2 + 0.3} \cdot (0.32\sigma_1 + 0.6),$$

$$b = 1 + \frac{\sigma_1\sigma_2^2}{10},$$

$$c = \frac{-0.01 + (\sigma_1 + \sigma_2^2)^2}{0.02 + (\sigma_1 + \sigma_2^2)^2} \cdot \left(1 - \frac{(\sigma_1 - 1)(\sigma_2 - 1)}{2}\right),$$

$$d = \frac{0.05}{0.025 + (\sigma_1 + \sigma_2 - 2)^2},$$

$$(\sigma_1, \sigma_2) \in [-1, 1]^2\}.$$

Example: feedforward noise compensation

$$\varepsilon = 0.005 \quad \beta = 10^{-7} \quad \Rightarrow \quad N = 5427$$

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Example: feedforward noise compensation

$$\varepsilon = 0.005 \quad \beta = 10^{-7} \quad \Rightarrow \quad N = 5427$$

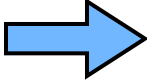
sample: $a_i, b_i, c_i, d_i \in \Delta, \quad i = 1, 2, \dots, 5427;$

solve:

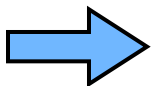
$$\min_{\nu_1, \nu_2} \left[\max_i \frac{(c_i + b_i \nu_1)^2 + (d_i + b_i \nu_2)^2 + 2a_i(c_i + b_i \nu_1)(d_i + b_i \nu_2)}{1 - a_i^2} \right]$$

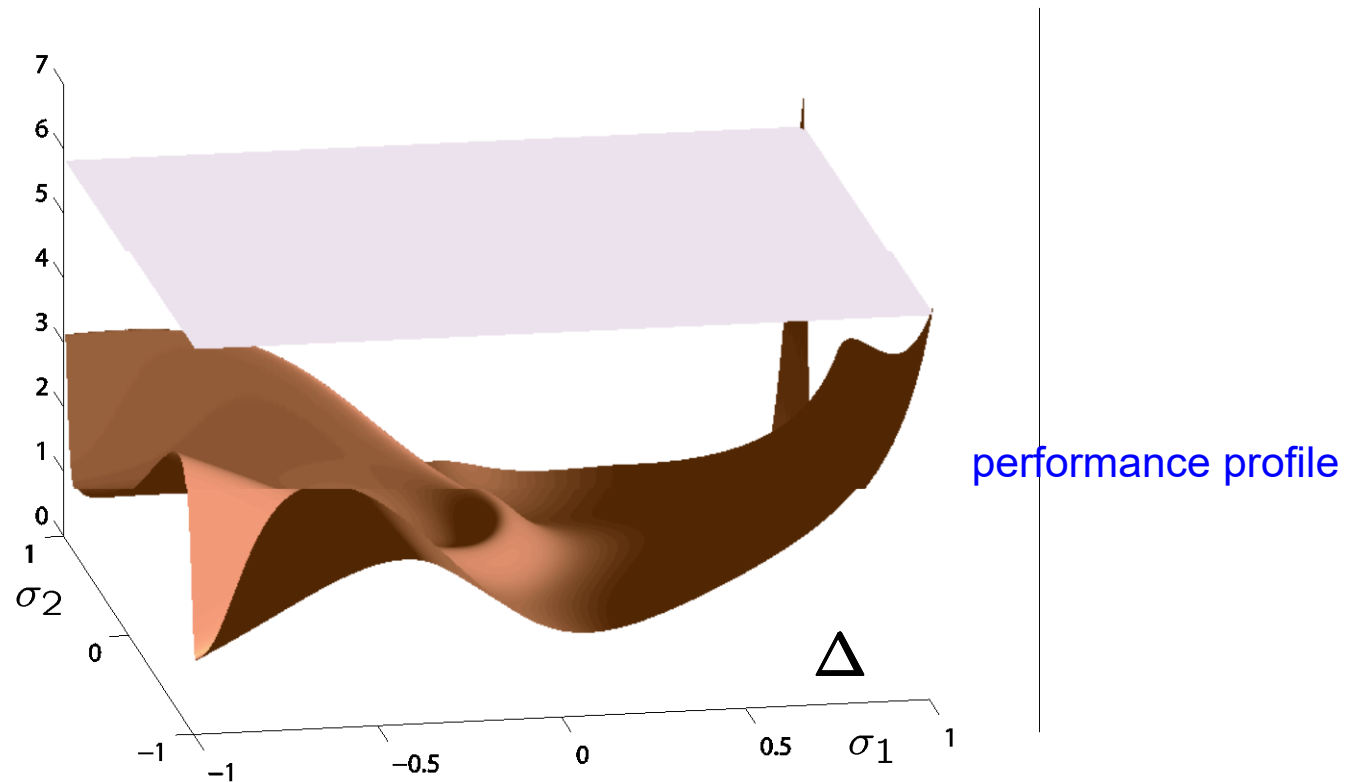
$$\Rightarrow \quad \nu_1^* = -0.9022, \quad \nu_2^* = -0.9028, \quad \ell^* = 5.8$$

Example: feedforward noise compensation

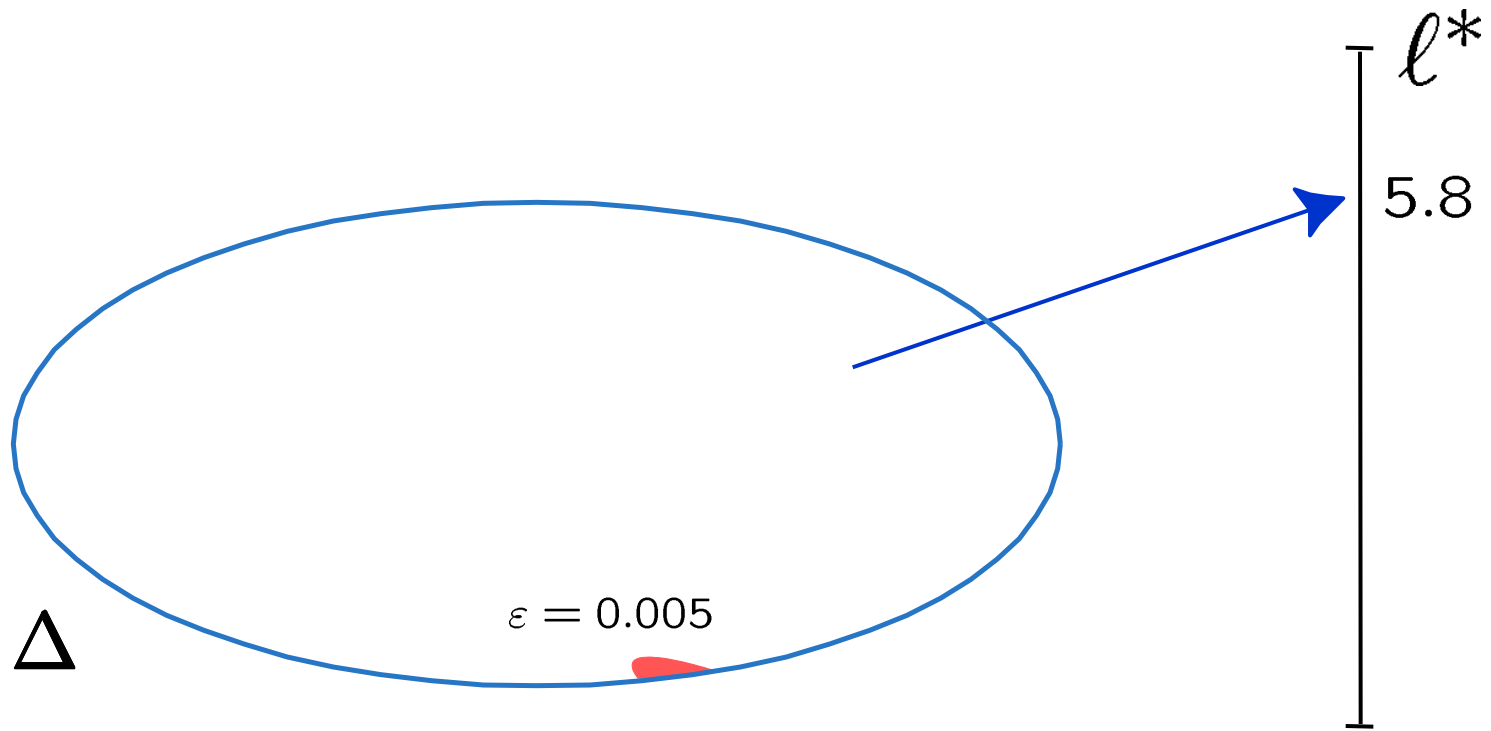
$\ell^* = 5.8$  **Output variance below 5.8 for all plants but a small fraction ($\varepsilon = 0.5\%$)**

Example: feedforward noise compensation

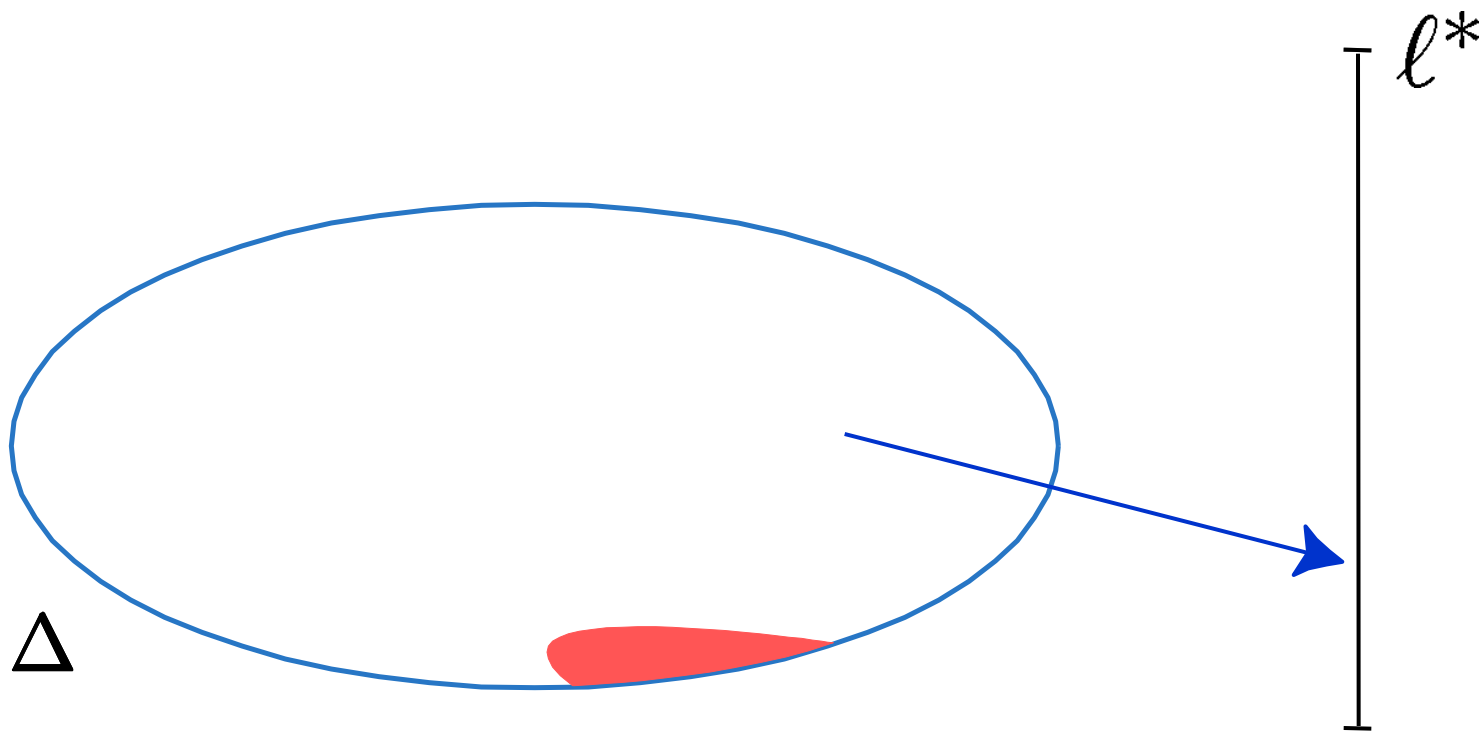
$\ell^* = 5.8$  **Output variance below 5.8 for all plants but a small fraction ($\varepsilon = 0.5\%$)**



Risk-Return Tradeoff



Risk-Return Tradeoff



Example: feedforward noise compensation

sample: $a_i, b_i, c_i, d_i \in \Delta, \quad i = 1, 2, \dots, 5427;$

solve:

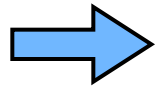
$$\min_{\nu_1, \nu_2} \left[\max_i \frac{(c_i + b_i \nu_1)^2 + (d_i + b_i \nu_2)^2 + 2a_i(c_i + b_i \nu_1)(d_i + b_i \nu_2)}{1 - a_i^2} \right]$$

Example: feedforward noise compensation

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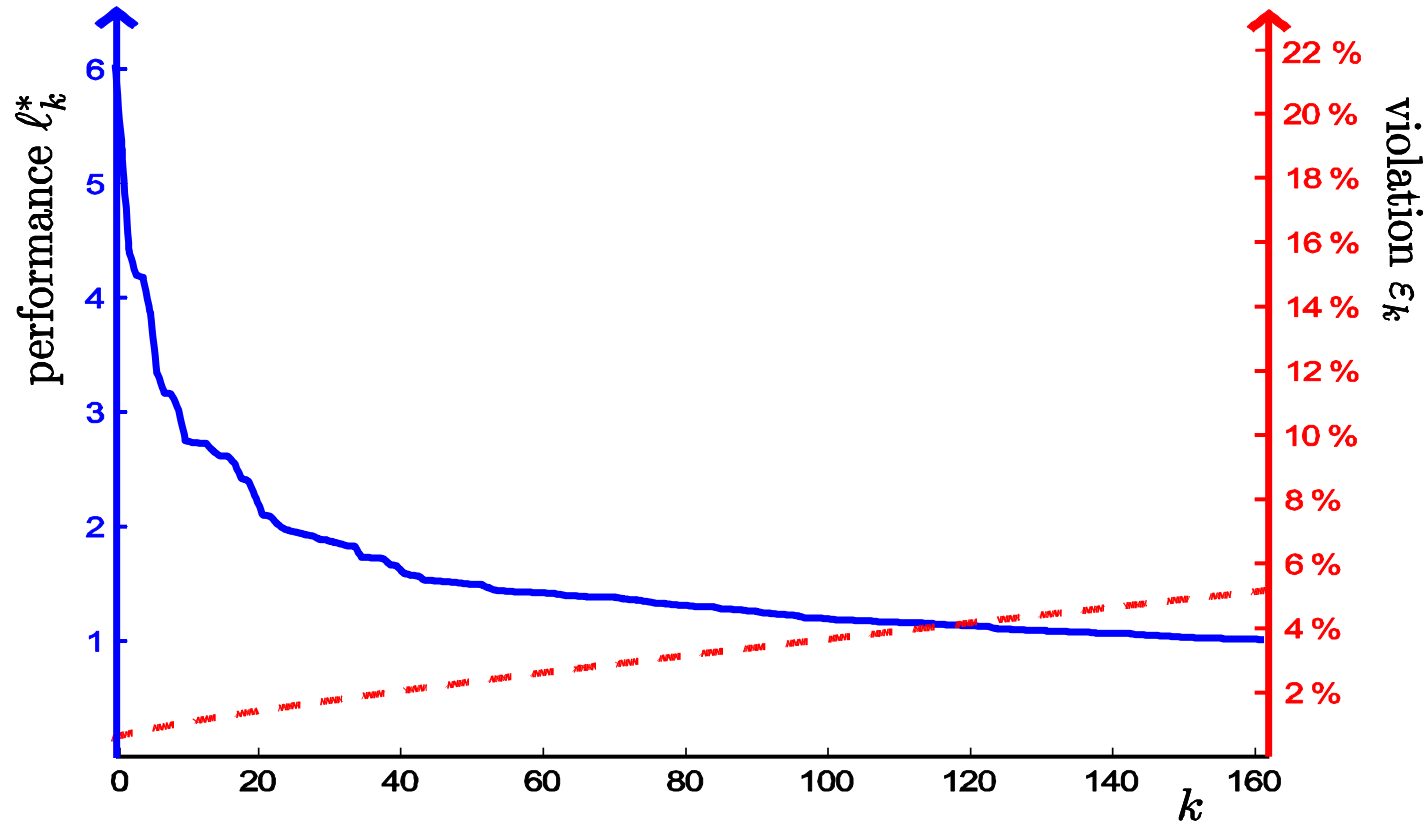
solve:

$$\min_{\nu_1, \nu_2} \left[\max_i \frac{(c_i + b_i \nu_1)^2 + (d_i + b_i \nu_2)^2 + 2a_i(c_i + b_i \nu_1)(d_i + b_i \nu_2)}{1 - a_i^2} \right]$$

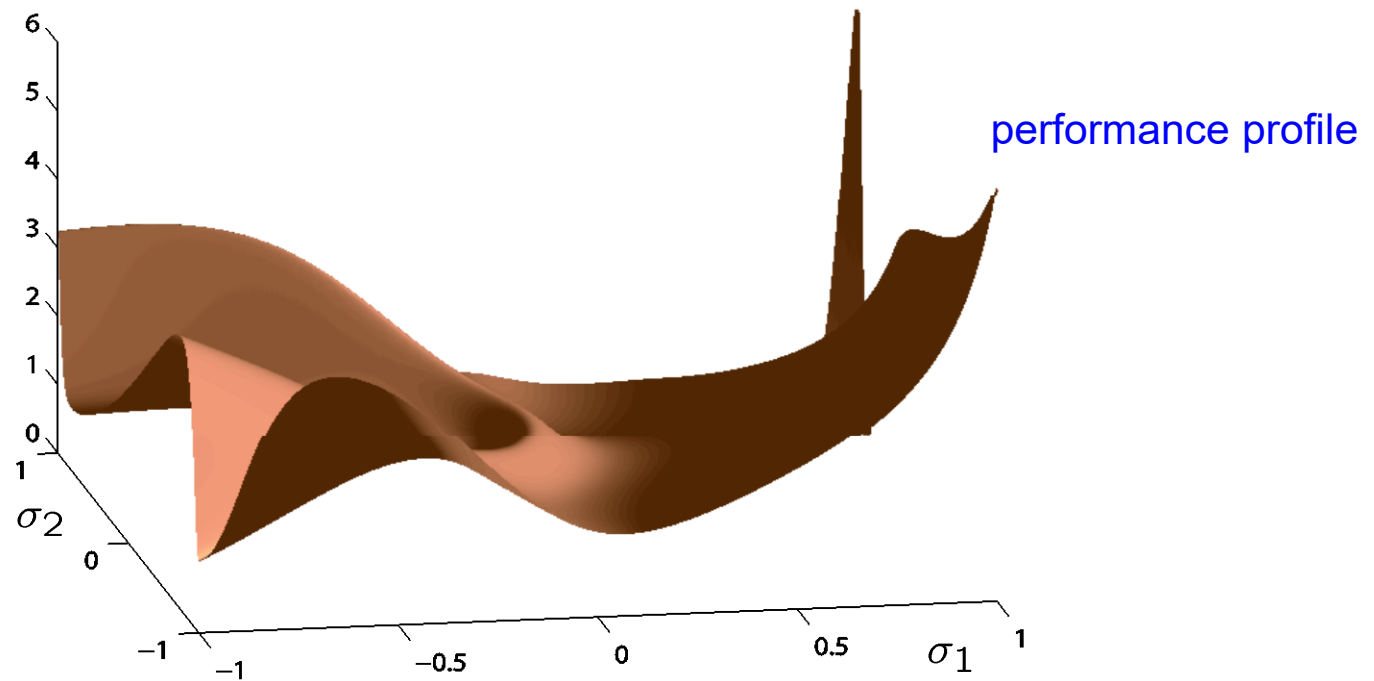


eliminate $k = 1, 2, \dots$ scenarios

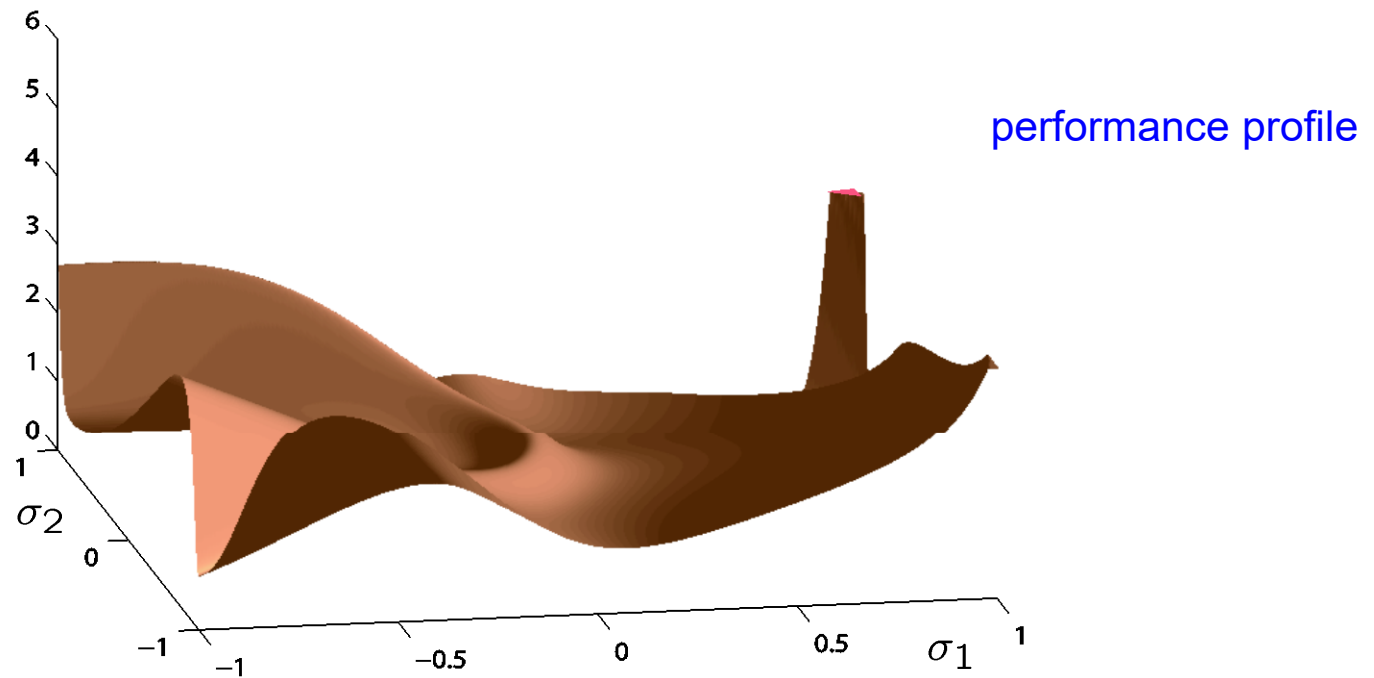
Example: feedforward noise compensation



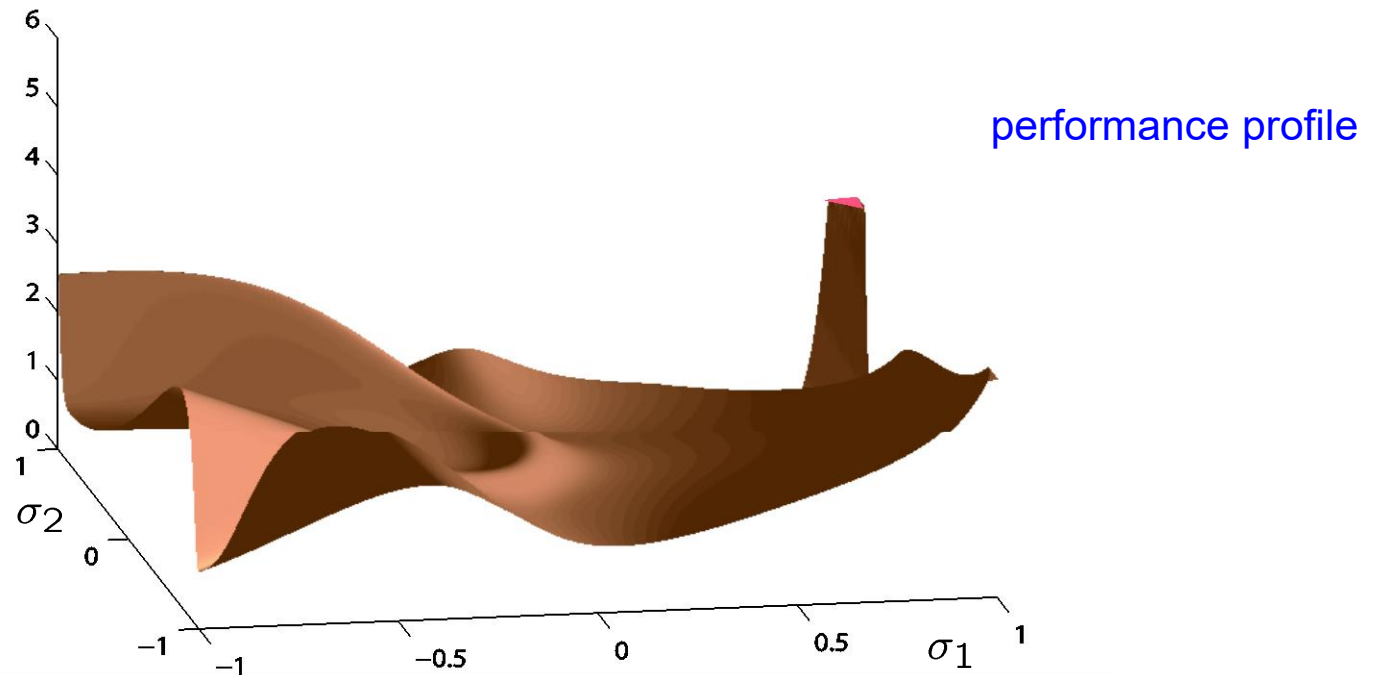
Example: feedforward noise compensation



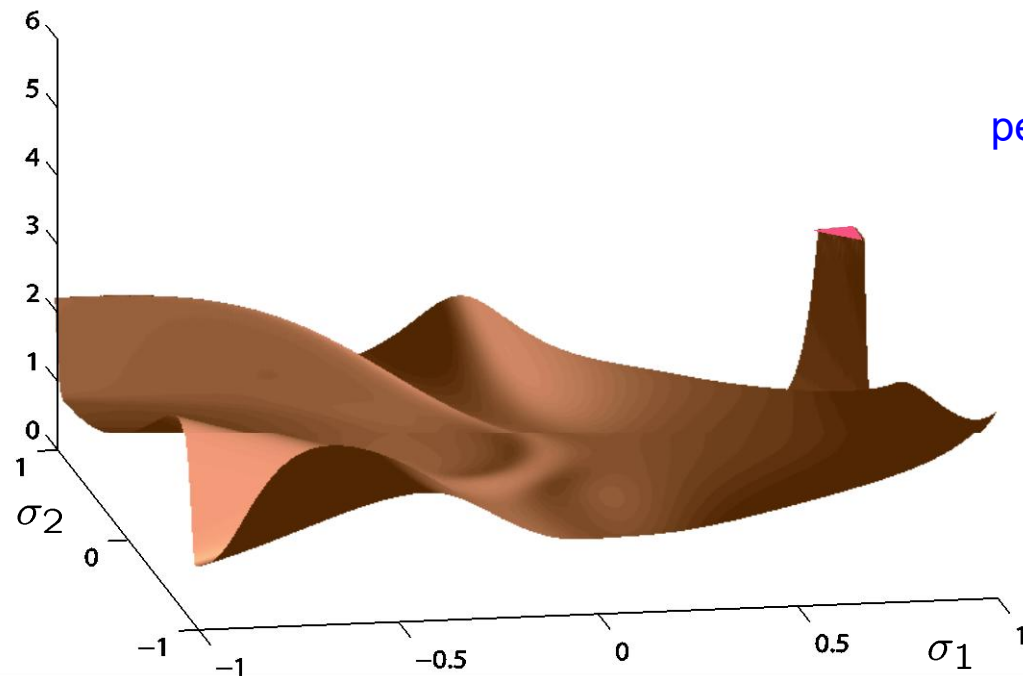
Example: feedforward noise compensation



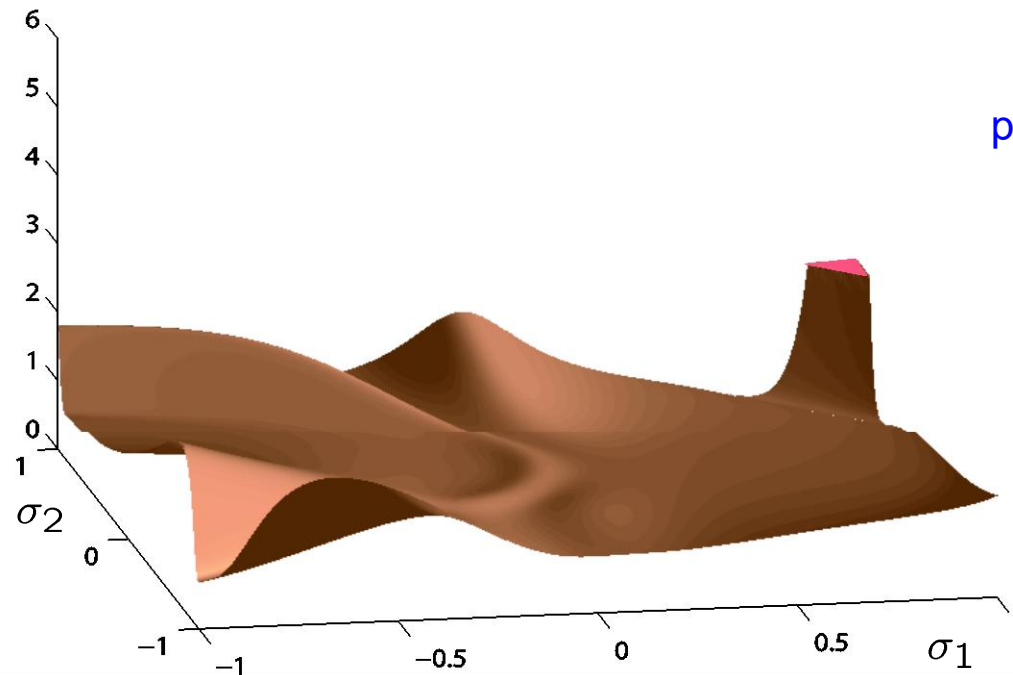
Example: feedforward noise compensation



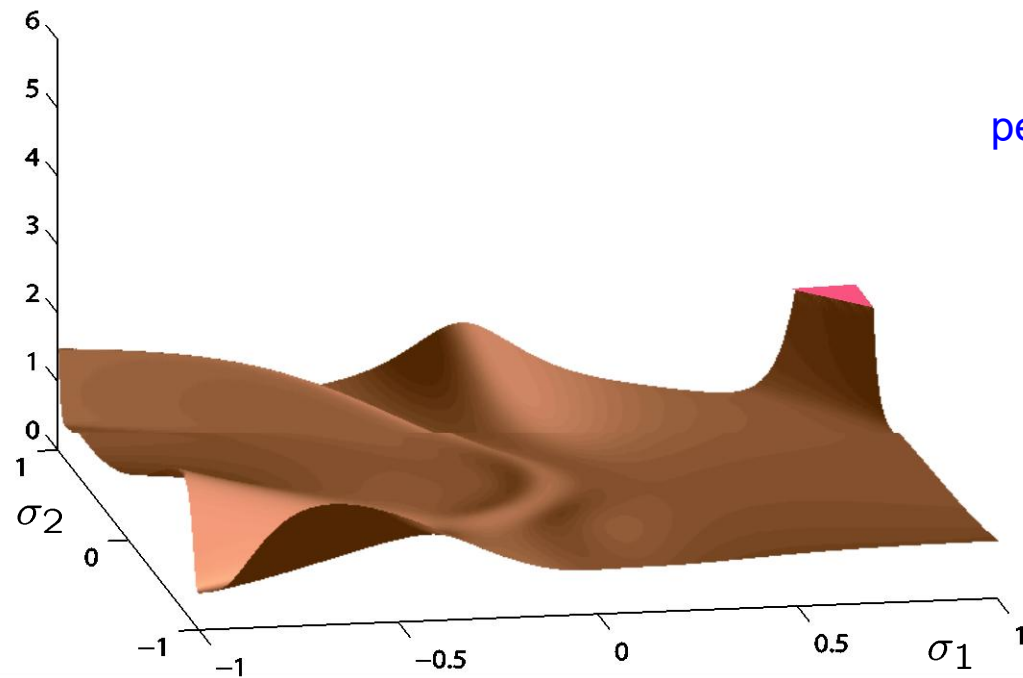
Example: feedforward noise compensation



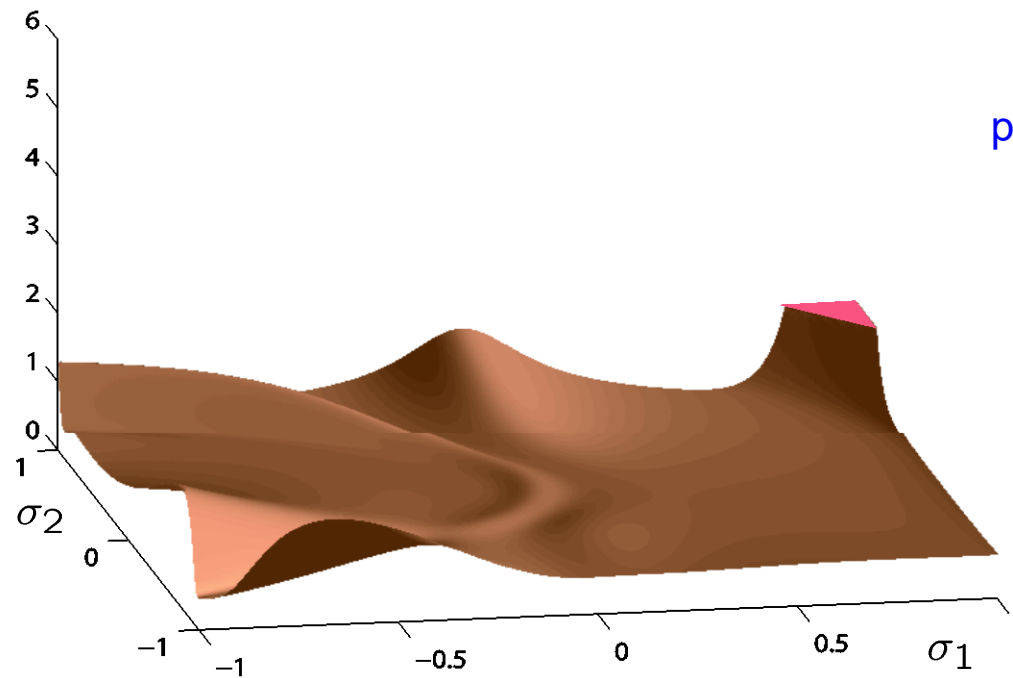
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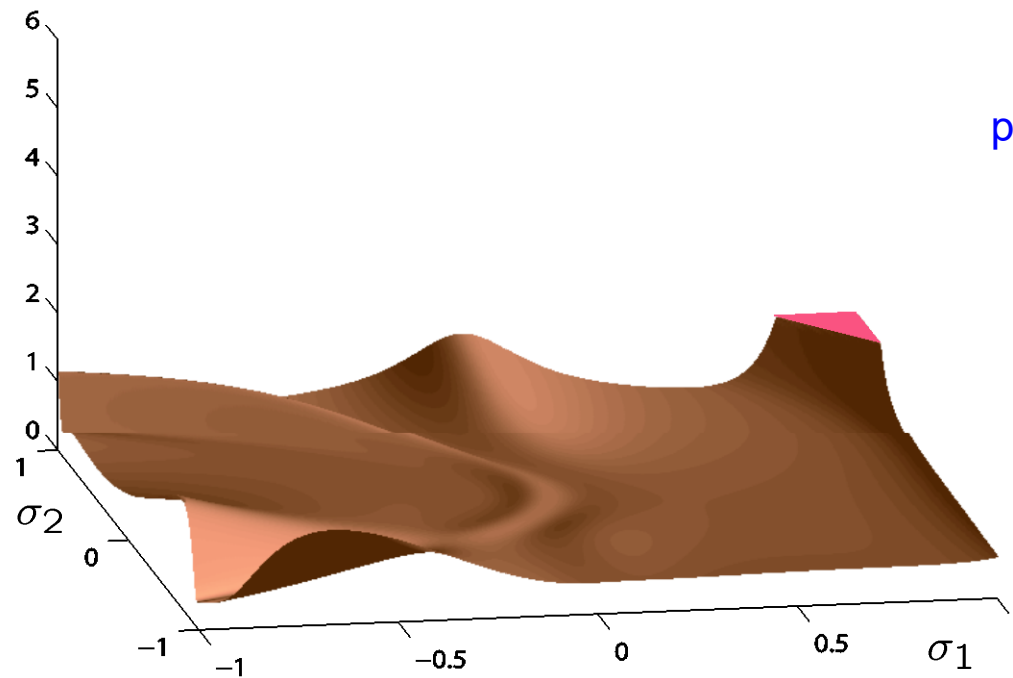
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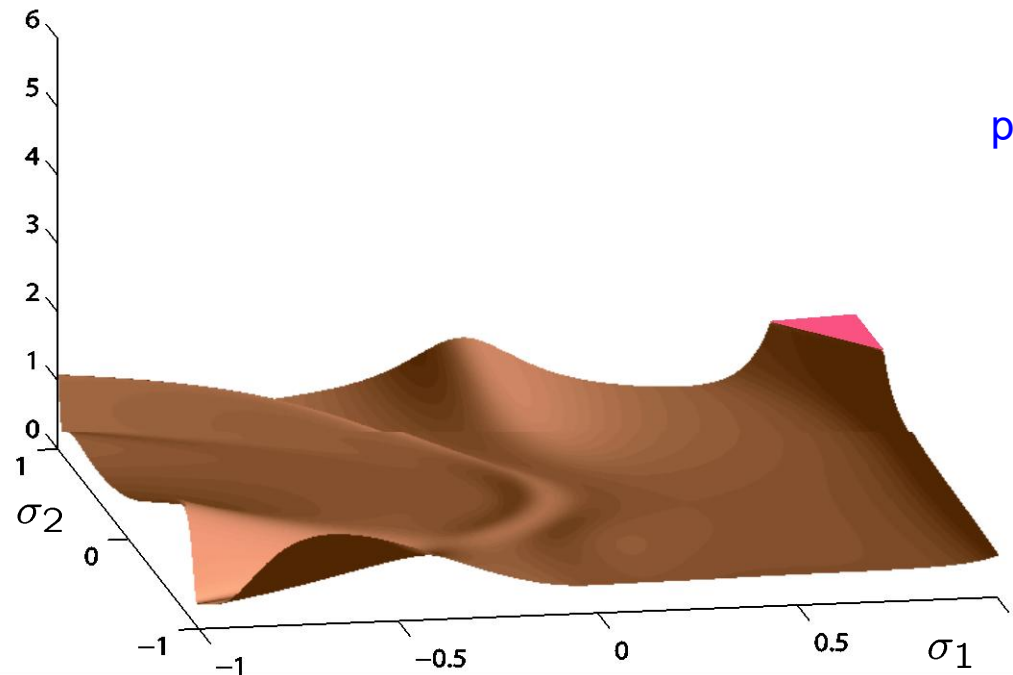


Example: feedforward noise compensation



performance profile

Example: feedforward noise compensation



performance profile

$$k = 60$$

$$l_{60}^* = 1.42$$

$$\epsilon_{60} = 2.5\%$$